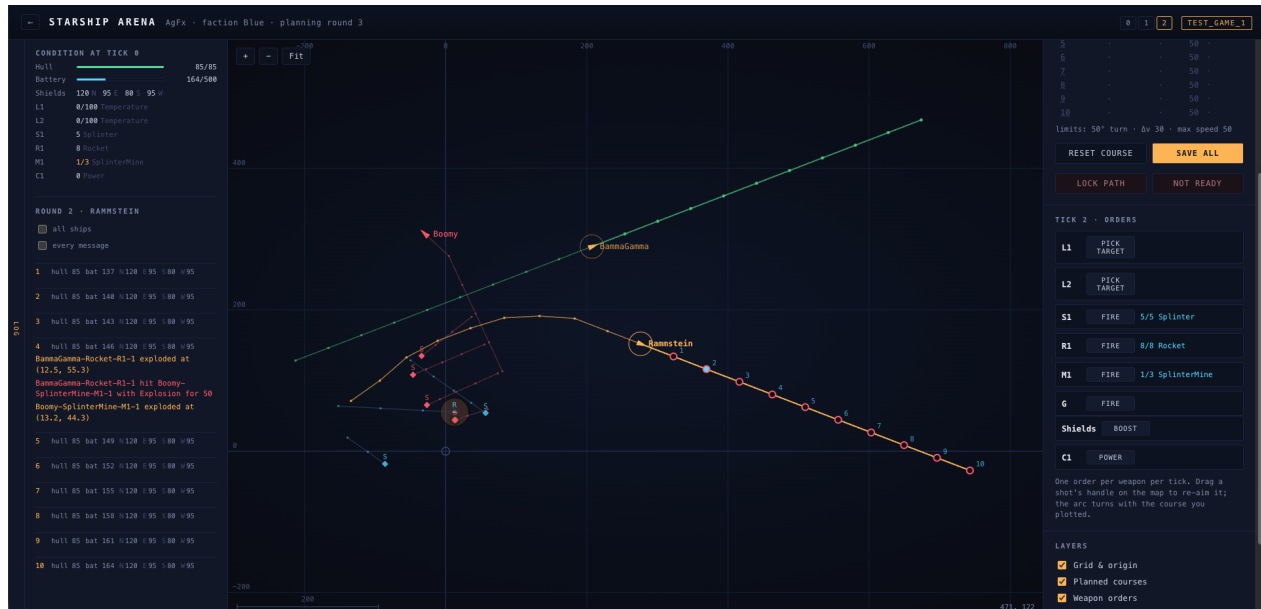


Starship Arena

Generated on 07 aug. 2026. The ship tables are read straight from the game, so they're what the engine will actually do.



The log on the left, the map in the middle, the orders for one tick on the right. Rammstein's course for the coming round is the orange chain, tick 2 is the joint picked out in blue, and that tick's weapons are listed beside it.

You command a starship. Often several.

It's a recreation of a play-by-mail game from 1991: teams of ships fighting each other in a 2D arena. It keeps the original's rhythm. You plan a round, everyone hands their orders in, the round runs, and you all find out together what happened.

A round is ten *ticks* of simulation. Nothing in it is random and nothing reads a clock. The same orders against the same starting position give you the same round, every time.

How a game runs

Two rounds matter at any moment: the last one that was played, and the one you're planning. The strip of numbers at the top of the map steps back through everything that's happened. Anything before the newest round is read only, and says so.

A round runs when the director processes it. Depending on the game, that's when the last player says Ready, or at fixed hours the game names.

A deadline beats readiness. If the hour comes round and your orders aren't in, nothing is entered for you and your ship carries on doing whatever it was already doing.

Your map

Three parts. The log folds out on the left, the map is in the middle, and the panel on the right is where you pick a ship and give it orders.

Drag to pan, scroll to zoom. **Fit** frames everything, including where you've told your ships to go. The grid spacing steps through 1, 2 and 5, and the bar in the bottom left

says what one square is worth. Your pointer's position in the world sits in the bottom right.

Plotting a course

Your course for the coming round is a chain of ten joints. Each segment is one tick: the direction it points is that tick's heading, its length is that tick's speed. Both carry forward until something changes them.

Grab a joint and drag. Everything after it swings along rigidly, because that's what the ship is going to do.

Tick 1 sits on top of the pile. At a standstill all ten joints are stacked on the ship, so the one you grab is the first tick, and pulling it gives the whole chain speed at once.

A joint turns red when your drag runs into a limit, and it stops there. The three limits are printed under the course table: degrees of turn per tick, change of speed per tick, and top speed.

Ticks that share a spot collapse into one label, 4-10. That's what a ship coasting to a stop looks like.

Lock path stops you dragging the course by accident while you fiddle with weapons. Weapons stay settable. **Reset course** puts it back to whatever the game has on file for you.

What you see while you drag is exact. A ship's own course follows only from its own orders, so the browser runs the same arithmetic the engine will. Anything involving another ship, like being hit, isn't predicted at all.

Orders for one tick

Click a joint on the course, or a tick number in the course table, and the panel below switches to that tick. Every weapon the ship carries is listed, and the button tells you what that weapon wants.

Button	What the weapon needs	How you give it
fire	A direction, in degrees relative to the ship's heading at that tick	Drag the handle on the map, or slide the control in the panel. Same number.
pick target	The name of something, which is what a laser takes	Click it on the map.
choose	One of a short list, such as a wreck for a starbase to replace	Pick from the dropdown.

While a tick is selected, each armed weapon's firing arc is drawn as a wedge, rotated to the heading the ship will have at *that* tick. Drag a shot outside the arc and it stops at the edge.

That's why a course and its shooting have to be planned together. The arc turns with the course you drew.

The gravscan's handle does two jobs at once. Turning it aims the sweep; pulling it further out narrows the cone. A narrow cone reaches much further, so the handle behaves the way the scanner does.

One order per weapon per tick. Ammunition is counted across your whole plan, so a launcher holding 6 rounds lets you plan 6 shots and then greys out.

Shields and cloaks sit under the weapons in the same panel. **boost** hands a shield quadrant energy, **power** sets what a cloak draws.

Saving, and Ready

Save all writes the orders for every ship you command. Save as often as you like: nothing's final until the round is processed.

Ready tells the director you're done thinking. The lamp beside each name under "Your ships" shows who's said it, yours and your faction's. In a game set up for it, the round is processed the moment the last player says Ready, and the map moves straight on.

They're separate on purpose. Saving keeps your work. Ready says stop waiting for me.

What you can see

Everything on your map came from a scan. The one exception is your own faction's ships, which you see as they really are.

Your picture is shared across the faction. Every scan every allied ship made during the round goes into it, so nobody has to send anybody screenshots.

A contact is a track of sightings joined up, and the newest point is where it was last seen. Its arrow comes from the last two sightings, so it means "it moved this way between these two moments". Something seen only once is a small diamond with no course at all.

A scan never gives away a target's real heading. The engine knows it and won't hand it over.

Terrain is the easy thing to spot. A rock is three times as visible as a ship the same size, so you'll see one long before it can matter.

The **Layers** switches turn off what you don't want to look at. A busy round can carry close to a hundred contacts, and most of them are somebody's ordnance in flight.

Reading the log

The **Log** tab on the left holds what happened, one block per tick. The row of figures is what the ship was down to at the end of that tick: hull, battery, and each shield quadrant.

Above it is the ship's condition as the round started, with every component showing what it has left against what a fresh one carries.

all ships widens the log to your whole faction. **every message** adds the internal chatter, and that's where a refused order turns up and says why. If an order did nothing and you can't see why, look there.

Movement

Everything in movement is per tick. A ship at speed 30 covers 30 units a tick, so 300 over a full round.

Three limits, all per tick. Turn is in degrees, so a ship that turns 30 comes all the way round in 12 ticks. Acceleration is in speed units, so one accelerating 20 towards a top speed of 40 gets there in 2 ticks. Negative speed is flying backwards, held to the same top speed.

Turning is always limited. Coming to a stop buys you nothing: a ship at rest turns no faster than one under way.

Movement costs energy, speed divided by 10 each tick and rounded down. A ship doing 46 spends 4 a tick. Let the battery fall below what your speed costs and the ship slows to what it can afford, which is worth knowing before you spend a round's energy on shields.

Turn and acceleration are both **relative**. They change the heading and speed you already have.

Energy

One battery per ship. The generators put their number into it every tick, and it stops at the maximum.

Spender	Cost
Movement	speed / 10 per tick, rounded down
Laser	5 per shot
Gravscan	10 per pulse
Cloak	whatever you set it to draw, every tick
Boosting a shield	1 energy per point of shield

A replenish fills the battery completely.

Weapons

Lasers

A laser is fired at something by name, and it hits whatever's in reach.

It has two numbers: the damage it does at contact, and the reach where that's fallen to nothing. In between it drops off *squared*, and that's what makes it the short range weapon. A laser of 200 reaching 100 does 200 at contact, 162 at 10 units, 50 at 50 units, and nothing at 100.

Both numbers are in the ship tables at the back, written Laser (200 to 100 (330, 30)): damage, reach, firing arc.

Every shot puts 20 heat in. It sheds 5 a tick and won't fire above 100, so firing flat out is something you can do for a while and not for a whole round.

A refused shot still costs you. If the laser's too hot, the battery's short, or the target's beyond what the ship can see, the heat and the energy go anyway. Don't order a shot every tick and hope.

Launchers

Point a launcher and it puts something into space, flying in the direction you gave it relative to the ship's heading, at its own speed rather than the ship's. Ammunition lasts the whole game unless you replenish.

A rocket flies straight. A guided missile looks for the nearest enemy inside a 45 degree cone out to 150 units, works out where that target is going, and slows on the approach so it closes the gap exactly. Guided missiles hunt other missiles too, so firing one back is a real defence.

Any damage at all destroys a missile. A 40 damage laser kills a Splinter exactly as dead as a 250 damage one, so coverage and rate of fire are what stop ordnance. Raw strength adds nothing.

Everything except the EMP missile flies at 60 and burns out after 15 ticks, so roughly 900 units of range. The EMP flies at 50, for 750.

Payload	Damage	Blast	What it does
Rocket	50	20	Unguided, and the full 50 anywhere inside the blast.
Splinter	75	6	Guided. Damage thins by 1 per unit, and the blast is only 6 across, so near enough 75 everywhere in it.
PowerSplinter	100	6	A heavier Splinter, and otherwise the same missile.
NanoMissile	100	50	Nanocytes eat matter. Double damage against hull, and nothing whatsoever against a shield with anything left in it. A wide blast for its class.
EMPMissile	100	10	Double against shields. Whatever's left over drains the battery instead of damaging the hull.

Mines

A mine leaves the tube at the ship's own speed less 5, sheds another 5 a tick, and stops. Drop one standing still and it stays exactly where you put it. That's the difference between laying a field and placing a mine where somebody's going to be.

A mine has 5 hull and enough battery for 50 ticks, five rounds, then it fizzles. Anything doing it more than 5 damage sets it off.

Splinter mines carry a Splinter warhead. Nanocyte mines carry a Splinter warhead *and* a nanocyte one, so they go off twice over.

The gravscan

A pulse costs 10 energy and sweeps a cone you point. The energy is spread over the cone, so a narrow pulse reaches much further than a wide one. The narrowest the order takes is 30 degrees.

A gravscan has a strength, and that's the reach of its narrowest pulse. It's the same kind of number as a hull's scan distance, so the two can be read against each other. Every gravscan in the game is a 200 today, which is 1200 units at 30 degrees:

Cone	30°	60°	90°	180°	360°
Reach	1200	848	692	489	346

Passive scanning runs from 156 to 330 depending on the hull, so looking everywhere at once is barely better than not pulsing at all, and a needle sees four times as far. Sweeping the whole circle in 30 degree pulses costs 120 energy against 10 for one wide look. Where you point it is a real decision.

A blast doesn't take sides

A warhead only goes off on an enemy. What it does once it's gone off doesn't care whose anything is.

Everything inside the radius takes the damage, including the ship that fired it, that ship's other launches, and its own mines. Fire the same launcher two ticks running and the second round is sitting inside the first one's blast when it goes off. Space your salvo, aim the halves apart, or accept the loss.

It cuts both ways in defence. Shooting an incoming missile down close to your own hull sets its warhead off exactly where it'll hurt you.

Defence

Shields

Shields are the outer defence: a quadrant has to break before anything reaches the hull. There are four, each covering 90 degrees, each with its own strength.

N is the front, 45 degrees either side of the bow, and they turn with the ship. Turning to present a stronger face genuinely works.

Each point of shield stops a point of damage. **Boost** moves battery into a quadrant one for one, up to twice that quadrant's normal maximum, and anything above the normal maximum drains away at the end of the round. A well timed boost lets you take a great deal.

Shields don't come back on their own. Boosting is how you rebuild one mid-fight, and replenishing at a starbase is the only thing that restores them outright. A shield you spent in round 2 is still spent in round 9.

Cloak

A cloak bends scans around the ship, as hard as the energy you put in. You set the draw with **power**, and it costs that much every tick until you change it.

Every cloak has a figure at which it *halves* an enemy's scan range. Twice that quarters it, three times and you're down to an eighth. A power of 0 is off and costs nothing, so there's no switch to remember.

The draw is capped at twice what the hull generates. Past that the curve has flattened and no battery pays for it long enough to matter.

A cloak running off spare generator output is a modest permanent advantage. Disappearing properly is a burst of a few rounds during which you're boosting no shields at all.

Terrain

Asteroids are solid, and running into one hurts.

You bounce. The part of your speed running along the surface is kept, and the part driving into it comes back at a third. A graze costs almost nothing. A square hit throws you back.

Damage is the speed you drove in with, so a ship flying straight into a rock at 45 loses 45, and it lands on the shield quadrant facing the rock. Below an impact speed of 5 there's no bounce at all, which is how a mine drifting into an asteroid settles against it instead of rattling about forever.

The bounce rewrites your heading and the helm gets no say. That's the real cost. You come out of it pointing somewhere you didn't choose, halfway through a round you plotted ten ticks of, and the rest of your course is fiction. Nobody ordered that turn, and that's exactly why it isn't held to your turn rate.

Learn the geometry and terrain becomes a tool. Clipping a rock brings you about faster than the helm allows, and it's a way to dump speed in a hurry.

A missile that hits terrain goes off against it, and a mine settles against it. Scans, lasers and blasts ignore terrain completely, so you can't yet hide behind a rock.

Replenishing

Get within 10 units of a starbase at a speed of 10 or less and give the **Replenish** order. It fills the hull, fills the battery, restores every shield quadrant, cools every laser and refills every magazine.

Scoring

Event	Score
Damaging a shield	1 point per 2 points of shield taken off
Breaking a shield to 0	25 points
Damaging a hull	1 point per point of hull, counting only the hull that was there
Draining a battery	1 point per 2 energy drained
The killing blow	100 points

Only the blow that finishes a ship scores the kill. Destroyed ships are cleared away at the end of the tick rather than the moment they die, so two gunners can both land a killing blow on the same missile without either wasting the shot, and a warhead going off later in the same tick still lands on a ship that's already dead.

Every order

Six things you can tell a ship to do, and every one of them belongs to a particular tick. Three address the ship itself. The rest address one of its components, which is why you pick the component before you set anything.

Order	Acts on	What it does
Accelerate	the ship	Changes speed for that tick, up or down, and far enough down it reverses you. Beyond the ship's limit it's refused outright. Dragging a joint sets it.
Turn	the ship	Turns so many degrees for that tick, either way. Beyond the ship's limit it's held to the limit and you're told. Dragging a joint sets this too.
Replenish	the ship	Restocks at a starbase you're close enough to and slow enough beside. Nothing to set.
Fire	one weapon	What it wants depends on the weapon: a direction for a launcher, a target for a laser, a direction and a cone width for a gravscan. One per weapon per tick.
Boost	the shields	Moves energy into one quadrant, 1 point of shield per point of energy.
Power	a cloak	Sets what the component draws every tick. It holds until you change it.

A refused order never fails quietly. It goes into the log with the reason, which is what **every message** is there to show you.

The tick, in order

Every tick runs the same phases in the same order, and knowing that order is worth something. Acceleration happens before turning, so slowing down and turning in the same tick gets you round faster than turning alone.

1. Anything due to arrive this round arrives.
2. Components do their upkeep. A laser sheds heat.
3. Generators make energy.
4. Components draw what they need per tick, and a ship that can't pay for its speed slows down to what it can afford.
5. Pre-move orders run: acceleration, then turning, then power and activation.
6. **Encounters resolve.** Every course is settled and nothing has moved yet, so this is where the tick works out what runs into what, and where warheads find what they go off on. Whatever happens earliest in the tick happens first, and everything after it is worked out against the world it left behind.
7. Everything travels whatever's left of its leg.
8. Post-move orders run: weapons fire, then boost and replenish. This is when new missiles and mines appear, and when laser damage is done.
9. Everything scans. Guided missiles pick a target and turn onto an intercept.
10. Movement energy is spent, mines slow down, missiles burn a little battery.
11. The tick is recorded, and anything destroyed is cleared away.

Ships

Every model in the game, with what it carries. All of it is public: nobody loses to a number they had no way of knowing.

Weapon entries read as damage to reach (firing arc) for a laser, (rounds payload (firing arc)) for a launcher and (strength) for a gravscan, where an arc of 360 points everywhere. Scan distance is the ship's own passive one, before any cloak.

A2545 Terrapin

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance
40	30	20	150	90/500	6	330
Weapons				Defence		
P1: Launcher (8 PowerSplinter (270, 90)) P2: Launcher (8 PowerSplinter (270, 90)) N1: Launcher (10 NanoMissile (270, 90)) R1: Launcher (10 Rocket (315, 45)) M1: Launcher (8 SplinterMine (360)) G: Gravscan (200)				Shield (120/115/105/115)		

A2527 Alligator

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance
45	35	25	105	90/500	7	252
Weapons				Defence		
L1: Laser (150 to 70 (300, 60)) P1: Launcher (6 PowerSplinter (300, 60)) N1: Launcher (6 NanoMissile (300, 60)) R1: Launcher (10 Rocket (315, 45)) M1: Launcher (8 SplinterMine (360)) G: Gravscan (200)				Shield (115/100/90/100)		

A2539 Caiman

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance
45	35	25	100	100/500	7	228
Weapons				Defence		
L1: Laser (140 to 75 (330, 30)) L2: Laser (110 to 65 (270, 90)) P1: Launcher (6 PowerSplinter (300, 60)) P2: Launcher (6 PowerSplinter (300, 60)) R1: Launcher (8 Rocket (315, 45)) M1: Launcher (8 SplinterMine (360)) G: Gravscan (200)				Shield (110/100/90/100)		
ECM						
C1: Cloak (halves at 5)						

A2553 Frog

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance
45	40	25	85	95/500	7	210
Weapons				Defence		
L1: Laser (160 to 65 (315, 45)) P1: Launcher (6 PowerSplinter (315, 45)) N1: Launcher (5 NanoMissile (315, 45)) R1: Launcher (8 Rocket (315, 45)) M1: Launcher (6 SplinterMine (360)) G: Gravscan (200)				Shield (105/90/80/90)		
ECM						
C1: Cloak (halves at 5)						

F2534 Cheetah

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance
60	50	30	70	95/500	8	156
Weapons				Defence		
L1: Laser (170 to 60 (330, 30)) R1: Launcher (10 Rocket (315, 45)) R2: Launcher (10 Rocket (315, 45)) M1: Launcher (3 SplinterMine (360)) G: Gravscan (200)				Shield (100/85/70/85)		
ECM						
C1: Cloak (halves at 6)						

F2551 Tiger

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance
50	50	30	85	100/500	8	180
Weapons				Defence		
L1: Laser (190 to 65 (330, 30)) L2: Laser (140 to 55 (300, 60)) S1: Launcher (5 Splinter (315, 45)) R1: Launcher (8 Rocket (315, 45)) M1: Launcher (3 SplinterMine (360)) G: Gravscan (200)				Shield (120/95/80/95)		
ECM						
C1: Cloak (halves at 6)						

F2547 Panther

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance
-------	------	--------------	------	---------	------------	---------------

50	45	25	85	95/500	7	192
Weapons				Defence		
L1: Laser (150 to 60 (315, 45)) S1: Launcher (6 Splinter (330, 30)) S2: Launcher (6 Splinter (330, 30)) N1: Launcher (5 NanoMissile (300, 60)) R1: Launcher (8 Rocket (315, 45)) M1: Launcher (4 SplinterMine (360)) G: Gravscan (200)				Shield (110/90/80/90)		
ECM						
C1: Cloak (halves at 6)						

F2533 Lion

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance
45	40	25	110	100/500	7	228
Weapons				Defence		
L1: Laser (160 to 70 (300, 60)) S1: Launcher (8 Splinter (30, 150)) S2: Launcher (8 Splinter (210, 330)) N1: Launcher (6 NanoMissile (315, 45)) R1: Launcher (10 Rocket (315, 45)) M1: Launcher (5 SplinterMine (360)) G: Gravscan (200)				Shield (130/105/90/105)		
ECM						
C1: Cloak (halves at 6)						

H2545 Cairo

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance
45	35	25	100	120/500	7	180
Weapons				Defence		
L1: Laser (150 to 70 (315, 45)) E1: Launcher (6 EMPMissile (300, 60)) R1: Launcher (10 Rocket (315, 45)) M1: Launcher (8 NanocyteMine (360)) G: Gravscan (200)				Shield (140/105/95/105)		

H2552 Babylon

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance
40	35	20	115	110/500	7	210
Weapons				Defence		
L1: Laser (160 to 70 (270, 90)) E1: Launcher (8 EMPMissile (300, 60)) S1: Launcher (8 Splinter (90, 270))				Shield (140/125/115/125)		

M1: Launcher (12 NanocyteMine (360)) G: Gravscan (200)	
ECM	
C1: Cloak (halves at 5)	

H2535 Rome

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance
30	30	15	155	80/500	6	300
Weapons				Defence		
SS1: Launcher (10 Splinter (30, 150)) SS2: Launcher (10 Splinter (30, 150)) SP1: Launcher (10 Splinter (210, 330)) SP2: Launcher (10 Splinter (210, 330)) N1: Launcher (8 NanoMissile (300, 60)) M1: Launcher (10 NanocyteMine (360)) G: Gravscan (200)				Shield (115/110/100/110)		

H2527 Athens

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance
45	40	20	100	110/500	8	210
Weapons				Defence		
L1: Laser (200 to 70 (330, 30)) L2: Laser (200 to 70 (330, 30)) E1: Launcher (4 EMPMissile (300, 60)) R1: Launcher (8 Rocket (315, 45)) G: Gravscan (200)				Shield (120/95/85/95)		

I2544 Hive

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance
30	25	15	150	80/500	7	270
Weapons				Defence		
L1: Laser (120 to 75 (300, 60)) SS1: Launcher (6 Splinter (30, 150)) SP1: Launcher (6 Splinter (210, 330)) R1: Launcher (8 Rocket (270, 90)) M1: Launcher (15 SplinterMine (360)) G: Gravscan (200)				Shield (145/140/125/140)		

I2552 Swarm

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance
25	20	10	185	85/500	6	240
Weapons				Defence		

R2551 Cobra

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance
40	40	25	85	120/500	8	156
Weapons				Defence		
L1: Laser (320 to 55 (345, 15)) L2: Laser (120 to 75 (315, 45)) R1: Launcher (5 Rocket (315, 45)) G: Gravscan (200)				Shield (120/75/60/75)		
ECM						
C1: Cloak (halves at 3)						

R2531 Dragon

Speed	Turn	Acceleration	Hull	Battery	Generators	Scan Distance
30	25	15	160	90/500	6	270
Weapons				Defence		
L1: Laser (240 to 60 (300, 60)) S1: Launcher (8 Splinter (30, 150)) S2: Launcher (8 Splinter (210, 330)) R1: Launcher (8 Rocket (315, 45)) G: Gravscan (200)				Shield (130/125/110/125)		
ECM						
C1: Cloak (halves at 4)						

Starbases

Starbases can't move, and they carry heavy weapons and deep shields to make up for it. Fly close enough and slow enough, give the Replenish order, and one will restock you.

SB2531

Replenish distance	Replenish speed	Hull	Battery	Generators	Scan Distance
10	10	250	200/1000	12	390
Weapons			Defence		
L1: Laser (300 to 60 (360)) L2: Laser (300 to 60 (360)) S1: Launcher (5 Splinter (360)) S2: Launcher (5 Splinter (360)) R1: Launcher (5 Rocket (360)) R2: Launcher (5 Rocket (360)) G: Gravscan (200) SS: Ship spawner (3)			Shield (400/400/400/400)		